

Fig. 6: An actual-size, single-side PCB for the power supply circuit

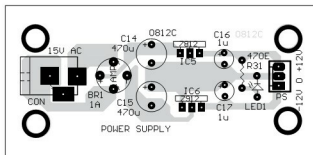


Fig. 7: Component layout for the PCB in Fig. 6

vides gain and level matching.

Potentiometers VR2 and VR6 control the frequency and amplitude. Using a log potentiometer as VR5 gives rather smoother control at low levels.

The dual-voltage ( $\pm 12V$ ) power supply required for the circuit is derived from AC mains. To derive the power supply, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 15V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising bridge rectifier BR1, filtered by capacitors C14 and C15, and regulated by ICs 7812 and 7912. Capacitors C16 and C17 bypass ripples in the regulated +12V and -12V.

## Construction

An actual-size, single-side PCB layout for the wave generator circuit is shown in Fig. 4 and its component layout in Fig. 5. Similarly, an actual-size, single-side PCB layout for the power supply circuit is shown in Fig. 6 and its component layout in Fig. 7.

Assemble the wave generator circuit on the PCB shown in Fig. 4. Similarly, assemble the power supply circuit on the PCB shown in Fig. 6.

Use IC bases for all the ICs. Enclose both the assembled PCBs, including transformer X1, in a suitable cabinet.

On the PCBs, provide suitable connectors for switches S1 through S4, and potentiometers VR2, VR5 and VR6 so that these can be extended out from the cabinet. Do not put IC1 through IC4 in the sockets without proper checking of IC voltages and polarity of capacitors C14 and C15, as the reversal of any of these may cause

loud noises and acrid smoke. Connect secondary terminals of transformer X1 to the power supply PCB using short lengths of wire.

## Calibration

After assembling the circuit, calibrate the wave generator for proper setting. A multimeter or scopemeter is required to set up the unit. The scopemeter, if available, is more convenient to check the waveforms.

Connect the unit to the mains and switch-on. LED1 should glow and  $\pm 12V$  should appear on the relevant IC pins. If this doesn't happen, switch off power to the unit immediately and check the power supply circuit.

Now insert IC1 through IC4 and set all presets and potentiometers to mid position. Select position 3 through range-selector switch S2. Also select position 3 (for triangular wave) through mode-selector switch S3 and set S4 to 0 dB. Rotate potentiometer VR2 clockwise and VR5 fully anticlockwise. A DC voltage should appear on the voltmeter. Adjust VR4 to cancel it and shift switch S3 to position 2 for sine wave. Adjust VR3 to cancel any offset that appears on the scope.

## PARTS LIST

### Semiconductors:

|          |                                                       |
|----------|-------------------------------------------------------|
| IC1, IC2 | - LM13700 dual operational transconductance amplifier |
| IC3      | - NE5534 high-performance op-amp                      |
| IC4      | - TL082 dual JFET input op-amp                        |
| IC5      | - 7812, +12V regulator                                |
| IC6      | - 7912, -12V regulator                                |
| LED1     | - 5mm red LED                                         |
| BR1      | - 1A bridge rectifier                                 |

### Resistors (all $\frac{1}{4}$ -watt, $\pm 5$ per cent carbon):

|                  |                                          |
|------------------|------------------------------------------|
| R1, R6           | - 100-kilo-ohm                           |
| R2, R8, R12, R17 | - 4.7-kilo-ohm                           |
| R3, R5, R7       | - 12-kilo-ohm                            |
| R4, R10, R30     | - 10-kilo-ohm                            |
| R9, R15, R29     | - 2.7-kilo-ohm                           |
| R11              | - 1-kilo-ohm                             |
| R13, R14         | - 120-ohm                                |
| R16              | - 47-kilo-ohm                            |
| R18              | - 100-ohm                                |
| R19              | - 470-kilo-ohm                           |
| R20              | - 270-kilo-ohm                           |
| R21              | - 22-kilo-ohm                            |
| R22-R24, R26-R28 | - 82-kilo-ohm                            |
| R25              | - 1.8-kilo-ohm                           |
| R31              | - 470-ohm                                |
| VR1              | - 4.7-kilo-ohm cermet preset             |
| VR2, VR6         | - 10-kilo-ohm linear potentiometer       |
| VR3, VR4         | - 10-kilo-ohm cermet preset              |
| VR5              | - 100-kilo-ohm logarithmic potentiometer |

### Capacitors:

|                  |                                 |
|------------------|---------------------------------|
| C1, C6           | - 1 $\mu$ F, 25V electrolytic   |
| C2, C7, C12, C13 | - 0.1 $\mu$ F polyester         |
| C3, C8           | - 10nF polyester                |
| C4, C9           | - 1nF polyester                 |
| C5, C10          | - 100pF polyester               |
| C11              | - 22pF polyester                |
| C14, C15         | - 470 $\mu$ F, 35V electrolytic |
| C16, C17         | - 1 $\mu$ F, 25V electrolytic   |

### Miscellaneous:

|      |                                                    |
|------|----------------------------------------------------|
| S1   | - Mains on/off switch                              |
| S2   | - Two-pole, 6-way rotary switch                    |
| S3   | - Single-pole, 3-way rotary switch                 |
| S4   | - Single-pole, double-throw switch                 |
| X1   | - 230V primary to 15V, 500mA secondary transformer |
| CON1 | - 2-pin connector                                  |

Connect the scopemeter at the output indicated in the figure and tune VR1, VR2 and VR6 to set the maximum frequency and amplitude to 1 kHz and 1V peak-to-peak, respectively. With a scope connected to the output, square, sine and triangle waves should be visible and the amplitude should be variable with VR6. Now the wave generator is ready for use. ●

The author is a regular contributor to EFM